

**LIFEWS AgriShine(TM) GrowMeal(TM)**

# **PRIMARY 4-6 GARDEN INVESTIGATION JOURNAL**

**STUDENT EDITION**

DRAW OR MAP YOUR SCHOOL GARDEN

Name: \_\_\_\_\_

Class: \_\_\_\_\_ School: \_\_\_\_\_

**QUESTION -> MEASURE -> INVESTIGATE -> ANALYZE -> EXPLAIN**

## HOW TO WORK LIKE A GARDEN INVESTIGATOR

**QUESTION** - Ask something that can be answered with observations or measurements.

**MEASURE** - Choose an appropriate tool, unit and repeatable method.

**INVESTIGATE** - Compare conditions fairly and keep important factors consistent.

**ANALYZE** - Organize data, calculate where appropriate, and look for patterns or variation.

**EXPLAIN** - Use evidence to answer the question and identify limitations.

**Safety and data integrity: follow teacher instructions; use tools correctly; do not taste unknown plants or solutions; avoid disturbing pollinators; record what you actually observe, including unexpected results.**

## MY INVESTIGATION TOOLKIT

| Tool / method | What it measures | Unit / scale | Safety or accuracy note |
|---------------|------------------|--------------|-------------------------|
|               |                  |              |                         |
|               |                  |              |                         |
|               |                  |              |                         |
|               |                  |              |                         |
|               |                  |              |                         |

Common records may include cm, m, cm<sup>2</sup>, m<sup>2</sup>, mL, L, g, kg, degrees C, time, count, tally, pH and EC. Use only units appropriate to the instrument and task.

## FAIR TEST CHECKLIST

| Investigation part | My choice / description |
|--------------------|-------------------------|
|                    |                         |
|                    |                         |
|                    |                         |
|                    |                         |
|                    |                         |

Think about: question, treatment or comparison, response variable, controlled variables, replication, measurement method, units and safety.

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 01 - Raised Food Garden

**QUESTION:** How does bed position affect plant growth?

**METHOD:** Measure plant height at edge and center positions.

**MAIN MEASUREMENT:** Plant height (cm)

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Compare mean or typical heights and explain one possible cause.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 02 - Raised Food Garden

**QUESTION:** Does mulch change surface moisture?

**METHOD:** Compare teacher-approved soil-moisture observations in mulched and unmulched spots.

**MAIN MEASUREMENT:** Moisture score / observation

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Identify which condition retained more moisture.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

### INVESTIGATION 03 - Raised Food Garden

**QUESTION:** How uniform is plant spacing?

**METHOD:** Measure spacing between consecutive plants in a sample row.

**MAIN MEASUREMENT:** Spacing (cm)

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Find the smallest, largest and range.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 04 - Raised Food Garden

**QUESTION:** How quickly does water enter the bed?

**METHOD:** Use a simple teacher-led infiltration test at two locations.

**MAIN MEASUREMENT:** Time / infiltration observation

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Compare locations and suggest a soil-related explanation.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 05 - Raised Food Garden

**QUESTION:** How productive is the bed?

**METHOD:** Record bed area and harvest mass or count.

**MAIN MEASUREMENT:** Area and yield

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Calculate yield per unit area where appropriate.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## RAISED FOOD GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 06 - Geometry Garden

**QUESTION:** Which garden shape uses space efficiently?

**METHOD:** Measure dimensions of two bed shapes.

**MAIN MEASUREMENT:** Length, width, perimeter

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Compare usable area and boundary length.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 07 - Geometry Garden

**QUESTION:** How accurate is our scale drawing?

**METHOD:** Measure real dimensions and represent them with a chosen scale.

**MAIN MEASUREMENT:** Measured and drawing lengths

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Check whether the scale is consistent.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 08 - Geometry Garden

**QUESTION:** Where are right angles used?

**METHOD:** Use a square or teacher-approved angle check on garden structures.

**MAIN MEASUREMENT:** Angle classification

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain why right angles may help construction.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 09 - Geometry Garden

**QUESTION:** How can a complex bed be divided into simple shapes?

**METHOD:** Sketch a compound garden area and divide it into rectangles or triangles.

**MAIN MEASUREMENT:** Dimensions and area

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Estimate total area by adding parts.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 10 - Geometry Garden

**QUESTION:** Can we redesign the layout?

**METHOD:** Create a revised garden plan within fixed dimensions.

**MAIN MEASUREMENT:** Area, path width, bed dimensions

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain one design trade-off.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## GEOMETRY GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 11 - Grid Experiment Garden

**QUESTION:** Does one treatment affect growth?

**METHOD:** Define a treatment and control and measure the same response in each.

**MAIN MEASUREMENT:** Treatment, control, response

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Compare results using evidence.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 12 - Grid Experiment Garden

**QUESTION:** Why do experiments need replication?

**METHOD:** Collect the same response from several grid cells per treatment.

**MAIN MEASUREMENT:** Repeated measurements

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Compare variation within treatments.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 13 - Grid Experiment Garden

**QUESTION:** How should treatments be assigned?

**METHOD:** Create a simple randomized treatment map with teacher guidance.

**MAIN MEASUREMENT:** Grid cell and treatment

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain how randomization improves fairness.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 14 - Grid Experiment Garden

**QUESTION:** What should stay constant?

**METHOD:** List variables that should not change during the investigation.

**MAIN MEASUREMENT:** Controlled variables

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain why one constant matters.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 15 - Grid Experiment Garden

**QUESTION:** What conclusion does the grid support?

**METHOD:** Summarize measurements from treatment and control cells.

**MAIN MEASUREMENT:** Mean/typical result and range

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** State a conclusion and one limitation.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## GRID EXPERIMENT GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 16 - Keyhole Garden

**QUESTION:** How far can a learner reach into the bed?

**METHOD:** Measure reach distance from the access path to planting zones.

**MAIN MEASUREMENT:** Distance (cm)

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Assess accessibility.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 17 - Keyhole Garden

**QUESTION:** How does position relate to moisture?

**METHOD:** Compare moisture observations near and far from the center basket.

**MAIN MEASUREMENT:** Moisture score

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Describe the spatial pattern.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 18 - Keyhole Garden

**QUESTION:** How large is the growing area?

**METHOD:** Measure radius or dimensions and estimate bed area.

**MAIN MEASUREMENT:** Length and area

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain assumptions in the estimate.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 19 - Keyhole Garden

**QUESTION:** What happens to organic material in the center?

**METHOD:** Observe changes in teacher-approved compost material over time.

**MAIN MEASUREMENT:** Date, appearance, volume

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Describe decomposition evidence.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 20 - Keyhole Garden

**QUESTION:** How well does the keyhole system work?

**METHOD:** Use access, moisture and crop observations as indicators.

**MAIN MEASUREMENT:** Multiple indicators

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** State one strength and one improvement.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## KEYHOLE GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 21 - Compost Garden

**QUESTION:** How does compost temperature change?

**METHOD:** Measure pile temperature on repeated dates with teacher supervision.

**MAIN MEASUREMENT:** Temperature (C)

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Describe the trend.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 22 - Compost Garden

**QUESTION:** Is the compost too wet or too dry?

**METHOD:** Use a teacher-approved moisture assessment.

**MAIN MEASUREMENT:** Moisture category

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Relate moisture to decomposition.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 23 - Compost Garden

**QUESTION:** How does material size affect breakdown?

**METHOD:** Compare similar materials prepared at different particle sizes.

**MAIN MEASUREMENT:** Appearance / breakdown score

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain the observed difference.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 24 - Compost Garden

**QUESTION:** How does compost volume change?

**METHOD:** Estimate or measure pile dimensions over time.

**MAIN MEASUREMENT:** Dimensions / volume estimate

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Describe volume reduction.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 25 - Compost Garden

**QUESTION:** Is the compost mature?

**METHOD:** Use agreed maturity indicators such as smell, texture and visible feedstock.

**MAIN MEASUREMENT:** Indicator checklist

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Give an evidence-based maturity judgment.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## COMPOST GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 26 - Gravity Drip Garden

**QUESTION:** How much water does an emitter deliver?

**METHOD:** Collect emitter discharge for a fixed time.

**MAIN MEASUREMENT:** Volume (mL), time

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Calculate flow rate.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 27 - Gravity Drip Garden

**QUESTION:** Are emitters delivering similar amounts?

**METHOD:** Measure discharge from several emitters.

**MAIN MEASUREMENT:** Emitter flow rates

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Compare variation and uniformity.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 28 - Gravity Drip Garden

**QUESTION:** How does tank height affect flow?

**METHOD:** Compare discharge at two teacher-approved water levels or heights.

**MAIN MEASUREMENT:** Height and flow

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Describe the relationship.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 29 - Gravity Drip Garden

**QUESTION:** Can we identify a clog or leak?

**METHOD:** Inspect the line and compare abnormal discharge with normal emitters.

**MAIN MEASUREMENT:** Flow / observation

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Diagnose the likely problem.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 30 - Gravity Drip Garden

**QUESTION:** How much water reaches the bed?

**METHOD:** Use emitter flow and irrigation duration.

**MAIN MEASUREMENT:** Flow rate and time

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Calculate total applied volume.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## GRAVITY DRIP GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 31 - Vertical Garden

**QUESTION:** Does plant growth differ by level?

**METHOD:** Measure comparable plants at upper, middle and lower levels.

**MAIN MEASUREMENT:** Height / leaf count

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Describe the vertical pattern.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 32 - Vertical Garden

**QUESTION:** How is light distributed?

**METHOD:** Use a simple light observation scale at different levels.

**MAIN MEASUREMENT:** Light category

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Relate light to growth.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 33 - Vertical Garden

**QUESTION:** How is water distributed?

**METHOD:** Observe moisture or drainage after watering at several levels.

**MAIN MEASUREMENT:** Moisture / drainage

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Identify uneven distribution.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 34 - Vertical Garden

**QUESTION:** How much floor space does the system save?

**METHOD:** Measure footprint and count productive levels.

**MAIN MEASUREMENT:** Footprint and levels

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain vertical space-use advantage.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 35 - Vertical Garden

**QUESTION:** Which level performs best?

**METHOD:** Use two indicators such as height and leaf count.

**MAIN MEASUREMENT:** Two performance indicators

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Justify the best-performing level.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## VERTICAL GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 36 - Crop Rotation Garden

**QUESTION:** Which crop families are in the rotation?

**METHOD:** Classify crops in the current and previous beds.

**MAIN MEASUREMENT:** Crop and family

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Identify repeated families.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 37 - Crop Rotation Garden

**QUESTION:** How long is the rotation interval?

**METHOD:** Use bed-history records to trace when a family returns.

**MAIN MEASUREMENT:** Years/terms between crops

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Evaluate whether repetition is frequent.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 38 - Crop Rotation Garden

**QUESTION:** Where can a legume fit?

**METHOD:** Design a sequence that includes a legume phase.

**MAIN MEASUREMENT:** Rotation sequence

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain the purpose of the legume phase.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 39 - Crop Rotation Garden

**QUESTION:** How does soil cover change?

**METHOD:** Compare ground cover after different crop phases.

**MAIN MEASUREMENT:** Cover estimate / category

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain implications for soil protection.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 40 - Crop Rotation Garden

**QUESTION:** Which rotation is more diverse?

**METHOD:** Compare two proposed sequences.

**MAIN MEASUREMENT:** Number of families / phases

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Recommend one and justify it.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## CROP ROTATION GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

# INVESTIGATION 41 - Water-Smart Garden

**QUESTION:** How fast does water enter the soil?

**METHOD:** Conduct a simple infiltration comparison at two locations.

**MAIN MEASUREMENT:** Time / infiltration rate

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain the difference.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 42 - Water-Smart Garden

**QUESTION:** Does mulch reduce surface drying?

**METHOD:** Compare teacher-approved moisture observations in mulched and bare soil.

**MAIN MEASUREMENT:** Moisture score

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Interpret the result.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

# INVESTIGATION 43 - Water-Smart Garden

**QUESTION:** How much water is applied?

**METHOD:** Measure container or irrigation volume used for a defined area.

**MAIN MEASUREMENT:** Volume and area

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Calculate water applied per area.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 44 - Water-Smart Garden

**QUESTION:** Where does runoff occur?

**METHOD:** Map visible runoff pathways during a supervised watering event.

**MAIN MEASUREMENT:** Location / observation

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Suggest one prevention measure.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

# INVESTIGATION 45 - Water-Smart Garden

**QUESTION:** How productive is the water used?

**METHOD:** Relate harvest quantity to measured irrigation water.

**MAIN MEASUREMENT:** Yield and water volume

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Calculate a simple water-productivity indicator.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## WATER-SMART GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

## SYSTEM MAP / FIRST OBSERVATION

Sketch the garden system. Label important structures, inputs, water pathways, crop zones or measurement points.

MY SYSTEM MAP

One investigation question I could ask: \_\_\_\_\_

## INVESTIGATION 46 - Intro Soilless Garden

**QUESTION:** How do growing media differ?

**METHOD:** Compare two media for texture, drainage and water retention.

**MAIN MEASUREMENT:** Observation scores

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Recommend a use based on evidence.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 47 - Intro Soilless Garden

**QUESTION:** What is the reservoir pH?

**METHOD:** Measure pH with a teacher-approved meter or strip.

**MAIN MEASUREMENT:** pH

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Interpret whether the reading changed over time.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 48 - Intro Soilless Garden

**QUESTION:** What does electrical conductivity tell us?

**METHOD:** Measure EC with teacher supervision.

**MAIN MEASUREMENT:** EC reading

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Explain that EC indicates dissolved-ion concentration, not individual nutrients.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 49 - Intro Soilless Garden

**QUESTION:** How does solution move through the system?

**METHOD:** Trace the water/nutrient-solution pathway.

**MAIN MEASUREMENT:** Flow path and observations

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** Identify one possible failure point.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INVESTIGATION 50 - Intro Soilless Garden

**QUESTION:** How is the root zone performing?

**METHOD:** Observe roots and plant condition without damaging the system.

**MAIN MEASUREMENT:** Root color/condition, plant signs

| Sample / position | Measurement 1 | Measurement 2 | Notes |
|-------------------|---------------|---------------|-------|
|                   |               |               |       |
|                   |               |               |       |
|                   |               |               |       |

GRAPH / DIAGRAM / CALCULATION

**ANALYZE:** State one evidence-based system-care recommendation.

**EXPLAIN:** My evidence supports the conclusion that \_\_\_\_\_

One limitation or source of uncertainty: \_\_\_\_\_

Next investigation or improvement: \_\_\_\_\_

Date: \_\_\_\_\_ Group: \_\_\_\_\_ Teacher initials: \_\_\_\_\_

## INTRO SOILLESS GARDEN - INVESTIGATION REVIEW

Select the strongest evidence you collected in this garden.

BEST EVIDENCE / GRAPH / CALCULATION

What did the evidence show? \_\_\_\_\_

What would improve the investigation? \_\_\_\_\_

What management or design decision could use this evidence? \_\_\_\_\_

MY LONGITUDINAL GROWTH DATA

Track the same crop or sample location on at least five dates using the same measurement method.

| Date | Height / response | Leaf / count metric | Condition / treatment | Notes |
|------|-------------------|---------------------|-----------------------|-------|
|      |                   |                     |                       |       |
|      |                   |                     |                       |       |
|      |                   |                     |                       |       |
|      |                   |                     |                       |       |
|      |                   |                     |                       |       |

PLOT THE CHANGE OVER TIME

MY DATA ANALYSIS PAGE

Choose one investigation and summarize its data. Calculate a mean, range, percentage, rate, area, flow, yield or other appropriate Primary 4-6 metric.

| Quantity | Equation / method | Working | Result + unit |
|----------|-------------------|---------|---------------|
|          |                   |         |               |
|          |                   |         |               |
|          |                   |         |               |
|          |                   |         |               |

GRAPH OR COMPARISON

## MY FINAL GARDEN INVESTIGATION

Question: \_\_\_\_\_

Prediction: \_\_\_\_\_

What I will measure: \_\_\_\_\_

| Sample | Measurement | Unit | Observation |
|--------|-------------|------|-------------|
|        |             |      |             |
|        |             |      |             |
|        |             |      |             |
|        |             |      |             |

Conclusion from evidence: \_\_\_\_\_

Limitation: \_\_\_\_\_

Next step: \_\_\_\_\_

## MY GROWMEAL(TM) PRIMARY 4-6 CERTIFICATE

This investigation journal belongs to

---

who completed garden investigations using

**QUESTION -> MEASURE -> INVESTIGATE -> ANALYZE -> EXPLAIN**

Teacher: \_\_\_\_\_ Date: \_\_\_\_\_

LIFEWS AgriShine(TM) GrowMeal(TM) - Primary 4-6